Mengran **Li**

PhD Candidate in Statistics

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Summary

I develop causal inference methodology designed for rare and extreme events. My work contributes to causal discovery, estimation, and attribution in settings involving extreme events and extremal dependence. By synthesizing causal frameworks with extreme value theory, I aim to enable reliable inference in high-impact extreme regimes with a focus on climate risk assessment.

Education

University of Glasgow — *Glasgow, UK*
Doctor of Philosophy in Statistics — *2022 - 2026*

University of Glasgow — *Glasgow, UK*
Master of Science in Statistics with distinction — *2021 - 2022*

Southwestern University of Finance and Economics — *Chengdu, China*
Bachelor of Science in Statistics (Major)
Bachelor of Science in Finance (Minor) — *2015 - 2019*

Publications

* denotes corresponding author

1. **Li, M.*** & Castro-Camilo, D. (2026) Causal Discovery in Multivariate Extremes via Tail Asymmetry. *arXiv:2604.21620.* Under review at *JASA* (Theory & Methods).
2. **Li, M.*** & Castro-Camilo, D. (2026). Tail-Calibrated Estimation of Extreme Quantile Treatment Effects. *arXiv:2603.23309.* Under review at *JRSS-B*.
3. **Li, M.*** & Castro-Camilo, D. (2025). On the importance of tail assumptions in climate extreme event attribution. *arXiv:2507.14019. Revision invited at Climatic Change.*
4. **Li, M.***, Cuba, D., Hu, C., & Castro-Camilo, D. (2024). A wee exploration of techniques for risk assessments of extreme events. *Extremes*. doi:10.1007/s10687-024-00500-5
5. **Li, M.*** & Castro-Camilo, D. (2025). eFCM: An R package for the Exponential Factor Copula Model. CRAN R package. cran.r-project.org/package=eFCM

Research Experience

Causal Structure Learning for Multivariate Extremes — *Glasgow, UK*
Developed structure learning methods for high-dimensional multivariate extremes and tail dependence, enabling recovery of directional causal structure encoded by directed acyclic graphs in extreme regimes. — *2025 - 2026*

Causal Inference for Extreme Quantile Treatment Effects — *Glasgow, UK*
Developed a causal inference framework based on inverse estimating equations for estimating treatment effects in the far tails of outcome distributions, with formal identification and asymptotic guarantees. — *2023 - 2025*

Tail Dependence and Causal Attribution of Climate Extremes — *Glasgow, UK*
Integrated multivariate extreme value modeling into a counterfactual causal inference framework to assess the role of tail dependence in extreme event attribution, showing that dependence assumptions critically shape causal conclusions. — *2022 - 2025*

Modeling Extreme Quantiles and Multivariate Extremes *Glasgow, UK*

Co-developed a modeling strategy for extreme quantiles and multivariate extremes, integrating extreme value and dependence modeling techniques. Results published in *Extremes*. *2023 - 2024*

Teaching Experience

University of Glasgow *Glasgow, UK*

Tutor and Demonstrator *2022 – 2026*

Statistics & Modeling	Bayesian Statistics, Time Series, Flexible Regression, Biostatistics, Advanced Predictive Models
Statistical Computing	Introduction to R & Python, Data Analysis
Responsibilities	Led tutorials and lab sessions for undergraduate and MSc students; provided technical support for programming assignments; assessed coursework and provided constructive feedback.

Conference Presentations

1. **First workshop of the Glasgow-Edinburgh Extremes Network**, 11 December 2025, University of Glasgow, Glasgow (UK). *Talk presentation.*
2. **ISI World Statistics Congress 2025**, 5–9 October 2025, International Statistical Institute, The Hague (Netherlands). *Talk presentation.*
3. **RSS 2025 International Conference**, 1–4 September 2025, Royal Statistical Society, Edinburgh (UK). *Poster presentation.*
4. **The 6th International Conference on Advances in Extreme Value Analysis and Application to Natural Hazards (EVAN 2024)**, 16–19 July 2024, Istituto Veneto di Scienze, Venice (Italy). *Talk presentation.*
5. **STOR-i Extremes Workshop**, 20–22 September 2023, Lancaster University, Lancaster (UK). *Talk presentation.*
6. **Research Students' Conference in Probability and Statistics**, 11–14 September 2023, University of Sheffield, Sheffield (UK). *Talk presentation; Travel grant recipient.*

Awards and Recognitions

- **CDSAI Training and Conference Fund (Competitive Award).** University of Glasgow, 2026.
- **Google Cloud Research Grant.** December 2025.
- **3rd Place, IASC Data Analysis Competition, Travel grant awarded.** ISI World Statistics Congress, The Hague (Netherlands), July 2025.
- **Fully funded PhD scholarship (CSC)**. 2022–2026.
- **Academic Scholarships**. Southwestern University of Finance and Economics. 2015–2019.

Academic Service

Journal Reviewer

Data-Centric Engineering (Cambridge University Press) *2025 - Present*

Seminar & Reading Group Organisation

Co-organized the monthly seminar series of the **Glasgow–Edinburgh Extremes Network (GLE^{2}N)**, coordinating invited speakers and supporting seminar delivery for sessions hosted at the University of Glasgow. *2025 - Present*

Skills

Programming & Computing	R (advanced), Python, C++, Linux, Git, HPC, Cloud Computing (Google Cloud)
Scientific Software & Reproducibility	R package development (CRAN), R Markdown, LaTeX
Web & Documentation	HTML/CSS, Hugo